Integrale triple

- Să se calculeze următoarele integrale triple $\iiint_\Omega f(x,y,z)\,dx\,dy\,dz$:

1. Ω este mulțimea mărginită de planele $x=0$, $y=0$, $z=0$ și planul $\alpha: x+y+2z=2$, $f(x,y,z)=x+y+z$

$x=0 \Rightarrow$ planul (yOz)

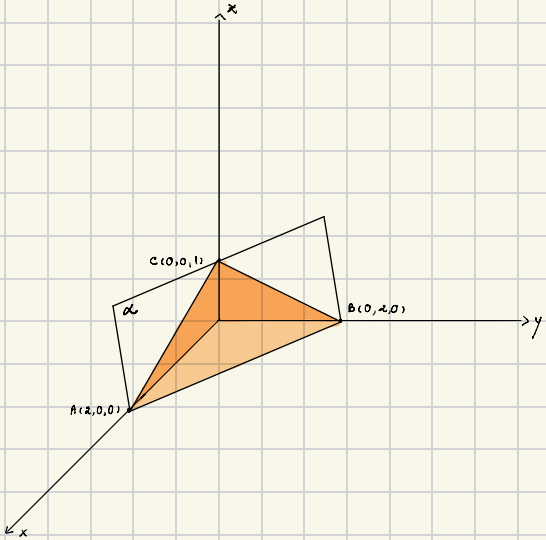
$y=0 \Rightarrow$ planul (xOz)

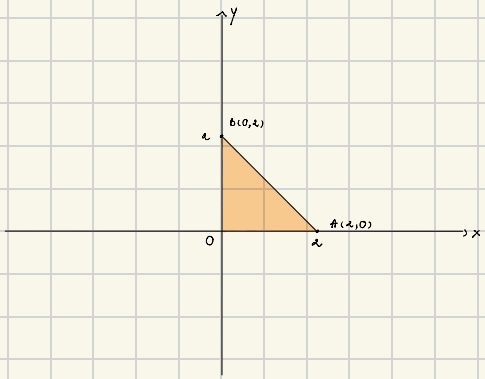
$z=0 \Rightarrow$ planul (xOy)

$\alpha \cap Ox: \; y=z=0 \Rightarrow x=2 \Rightarrow A(2,0,0)$

$\alpha \cap Oy: \; x=z=0 \Rightarrow y=2 \Rightarrow B(0,2,0)$

$\alpha \cap Oz: \; x=y=0 \Rightarrow z=1 \Rightarrow C(0,0,1)$





$0 \le z \le \frac{1}{2}(2-x-y)$

$$\iiint_\Omega f(x,y,z)\,dx\,dy\,dz = \iint_\Delta \left(\int_0^{\frac{1}{2}(2-x-y)} (x+y+z)\,dz \right) dx\,dy = -\frac{1}{8} \iint_\Delta (3x^2+6xy+3y^2-4x-4y-4)\,dx\,dy$$

$\Delta = pr_{xOy}\Omega = \triangle OAB = \{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 2,\ 0 \le y \le -x+2\}$

$AB: \; x+y=2 \iff AB: \; y=-x+2$

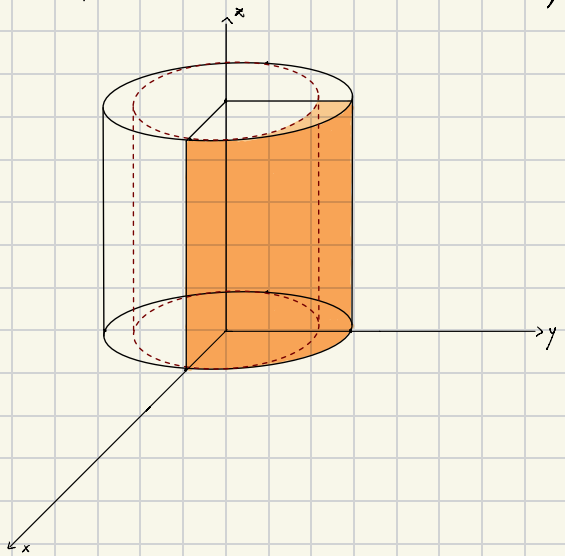
$$-\frac{1}{8} \iint_\Delta (3x^2+6xy+3y^2-4x-4y-4)\,dx\,dy = -\frac{1}{8} \int_0^2 \left(\int_0^{-x+2} (3x^2+6xy+3y^2-4x-4y-4)\,dy \right) dx =$$

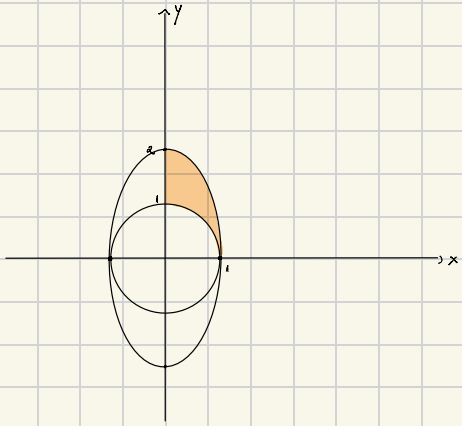
$$= -\frac{1}{8} \int_0^2 (-x^3+2x^2+4x-8)\,dx = \frac{5}{6}$$

2. $\Omega = \left\{ (x,y,z) \in \mathbb{R}^3 \;\middle|\; x^2 + \frac{y^2}{4} \le 1,\ x^2+y^2 \ge 1,\ x \ge 0,\ y \ge 0,\ 0 \le z \le 5 \right\}$, $f(x,y,z) = \frac{yz}{\sqrt{x^2+y^2}}$

$x^2 + \frac{y^2}{4} \le 1$ $\Rightarrow$ interiorul unui cilindru eliptic de semiaxe $a=1$ și $b=2$, cu axa de simetrie Oz

$x^2 + y^2 \ge 1$ $\Rightarrow$ exteriorul unui cilindru circular drept, cu secțiunea cerc de rază 1





$0 \le z \le 5$

$$\iiint_\Omega f(x,y,z)\,dx\,dy\,dz = \iint_D \left(\int_0^5 \frac{yz}{\sqrt{x^2+y^2}}\,dz \right) dx\,dy = \frac{25}{2} \iint_D \frac{y}{\sqrt{x^2+y^2}}\,dx\,dy$$

$D = pr_{xOy}\,\Omega = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; x^2 + \frac{y^2}{4} \le 1,\ x^2+y^2 \ge 1,\ x \ge 0,\ y \ge 0 \right\}$

$x^2 + \frac{y^2}{4} \le 1$ $\Rightarrow$ interiorul unei elipse de semiaxe $a=1$, $b=2$

$x^2 + y^2 \ge 1$ $\Rightarrow$ exteriorul unui cerc de centru $(0,0)$ și rază 1.

$D = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; 0 \le x \le 1,\ \sqrt{1-x^2} \le y \le 2\sqrt{1-x^2} \right\}$

$$\iint_D \frac{y}{\sqrt{x^2+y^2}}\,dx\,dy = \int_0^1 \left(\int_{\sqrt{1-x^2}}^{2\sqrt{1-x^2}} \frac{y}{\sqrt{x^2+y^2}}\,dy \right) dx = \frac{25}{2} \int_0^1 \left(\sqrt{x^2+y^2} \,\Big|_{\sqrt{1-x^2}}^{2\sqrt{1-x^2}} \right) dx = \frac{25}{2} \int_0^1 \left(\sqrt{-3x^2+4} - 1 \right) dx =$$

$$= \frac{25}{2} \int_0^1 \sqrt{4-3x^2}\,dx - \frac{25}{2} = \frac{25}{12\sqrt{3}} (4\widetilde{\pi} - 3\sqrt{3})$$

$$i = \int_0^1 \sqrt{4-3x^2}\,dx = \int_0^1 \frac{4-3x^2}{\sqrt{4-3x^2}}\,dx = \frac{4}{\sqrt{3}} \int_0^1 \frac{(\sqrt{3}x)'}{\sqrt{4-3x^2}}\,dx + \int_0^1 x \cdot \frac{-3x}{\sqrt{4-3x^2}}\,dx = \frac{4}{\sqrt{3}} \arcsin \frac{\sqrt{3}x}{2} \Big|_0^1 + \int_0^1 x \cdot (\sqrt{4-3x^2})'\,dx =$$

$$= \frac{4}{3\sqrt{3}}\widetilde{\pi} + 1 - \int_0^1 \sqrt{4-3x^2}\,dx \Rightarrow 2i = \frac{4}{3\sqrt{3}}\widetilde{\pi} + 1 \Rightarrow i = \frac{2}{3\sqrt{3}}\widetilde{\pi} + \frac{1}{2}$$

3. Ω este mulțimea mărginită de cilindrul $x^2 + y^2 = a^2$ și planele $z = 0$, $z = 1$, $y = x$, $y = \sqrt{3}x$, $f(x,y,z) = xy$

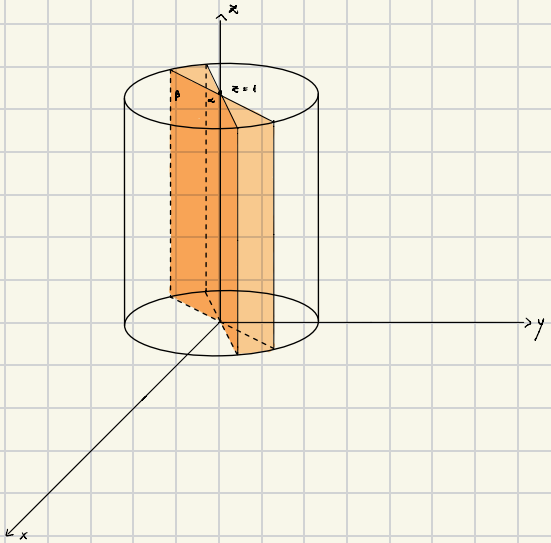
$z = 0 \Rightarrow$ planul (xOy)

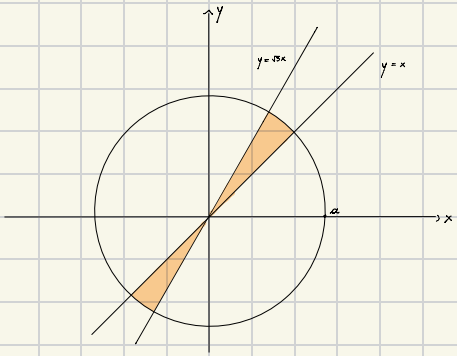
$z = 1 \Rightarrow$ plan paralel cu (xOy)

$y = x \Leftrightarrow x - y = 0 \Rightarrow$ plan care conține axa Oz (α)

$y = \sqrt{3}x \Leftrightarrow y - \sqrt{3}x = 0 \Rightarrow$ plan care conține axa Oz (β)

$x^2 + y^2 = a \Rightarrow$ cilindru circular drept, cu secțiunea cerc de rază a, axa de simetrie Oz





$0 \le z \le 1$

$$\iiint_\Omega f(x,y,z)\,dx\,dy\,dz = \iint_D \left(\int_0^1 xy\,dz \right) dx\,dy = \iint_D xy\,dx\,dy$$

$D = pr_{xOy}\Omega = \frac{1}{2}A$, unde A este sectorul de cerc de rază a, cuprins între dreptele de ecuație $y = x$ și $y = x\sqrt{3}$, pentru $x \ge 0$

$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r \in [0,a]$, $t \in \left[\frac{\pi}{4}, \frac{\pi}{3}\right]$

$$\iint_D xy\,dx\,dy = 2\int_{\pi/4}^{\pi/3} \left(\int_0^a r^3 \sin t \cos t\,dr \right) dt = \frac{1}{2}\int_{\pi/4}^{\pi/3} a^4 \sin t \cos t\,dt = \frac{a^4}{4}\int_{\pi/4}^{\pi/3} \sin 2t\,dt = -\frac{a^4}{8}\cos 2t \Big|_{\pi/4}^{\pi/3} =$$

$$= \frac{a^4}{8}(1 - 2\sin^2 t)\Big|_{\pi/4}^{\pi/3} = \frac{a^4}{8} \cdot \frac{1}{2} = \frac{a^4}{16}$$

- Să se calculeze volumul mulțimilor mărginite de suprafețele indicate

4. $z = x^2 + y^2$, $z = x + y$

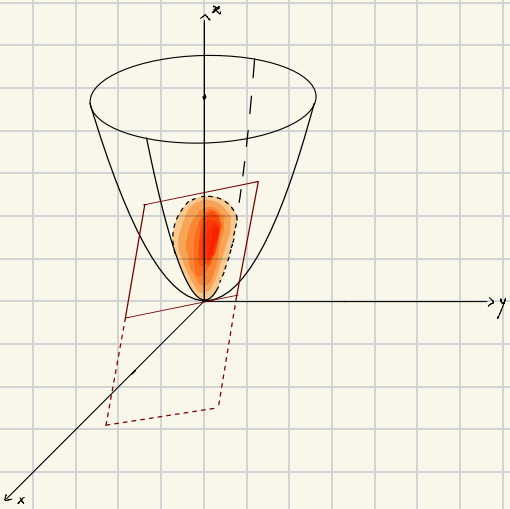
$z = x^2 + y^2 \Rightarrow$ paraboloid eliptic cu vârful în $O(0,0,0)$, desfășurare „în sus", axa de simetrie Oz, secțiune cerc

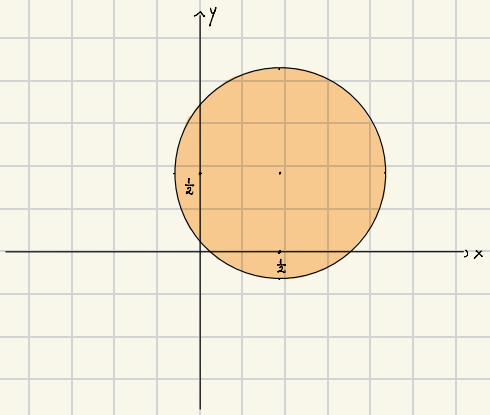
$z = x + y \Leftrightarrow x + y - z = 0 \Rightarrow$ plan ce conține originea (α)

$\alpha \cap Ox$: $z = y = 0 \Rightarrow x = 0 \Rightarrow O(0,0,0)$

$\alpha \cap Oy$: $x = z = 0 \Rightarrow y = 0 \Rightarrow O(0,0,0)$

$\alpha \cap Oz$: $x = y = 0 \Rightarrow z = 0 \Rightarrow O(0,0,0)$





$$x^2 + y^2 \le z \le x + y$$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_\Delta \left(\int_{x^2+y^2}^{x+y} dz \right) dx\,dy = \iint_\Delta (x + y - x^2 - y^2)\,dx\,dy$$

$x^2 + y^2 = x + y \Leftrightarrow x^2 - x + \frac{1}{4} + y^2 - y + \frac{1}{4} = \frac{1}{2} \Leftrightarrow \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \left(\frac{1}{\sqrt{2}}\right)^2 \Rightarrow$

$\Delta = pr_{xOy}\,\Omega$ = interiorul unui cerc de centru $\left(\frac{1}{2}, \frac{1}{2}\right)$ și rază $\frac{1}{\sqrt{2}}$

$$\Delta = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; \left(x - \tfrac{1}{2}\right)^2 + \left(y - \tfrac{1}{2}\right)^2 \le \left(\tfrac{1}{\sqrt{2}}\right)^2 \right\}$$

$x = r\cos t + \frac{1}{2}$

$y = r\sin t + \frac{1}{2}$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

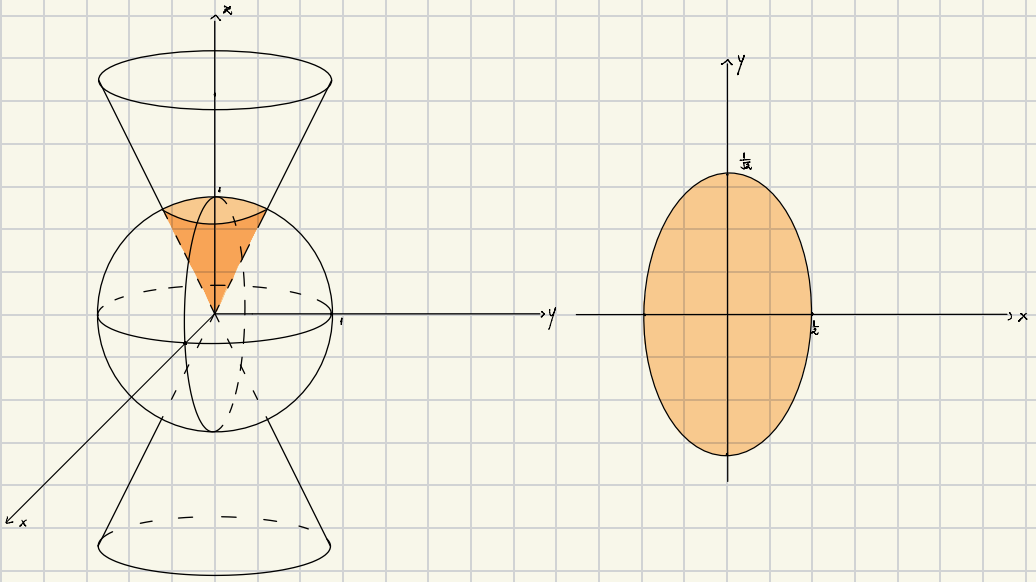
$r \in \left[0, \frac{1}{\sqrt{2}}\right]$, $t \in [0, 2\pi]$

$$\iint_\Delta (x + y - x^2 - y^2)\,dx\,dy = \int_0^{2\pi} \left(\int_0^{\frac{1}{\sqrt{2}}} (-r^2 + \tfrac{1}{2} r)\,dr \right) dt = \frac{1}{16} \int_0^{2\pi} dt = \frac{1}{16} \cdot 2\pi = \frac{\pi}{8} \Rightarrow V = \frac{\pi}{8}$$

5. $2x^2 + y^2 + z^2 = 1, \quad 2x^2 + y^2 - z^2 = 0, \quad z \geq 0$

$2x^2 + y^2 + z^2 = 1 \iff \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + y^2 + z^2 - 1 = 0 \Rightarrow$ elipsoid

$2x^2 + y^2 - z^2 = 0 \iff \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + y^2 - z^2 = 0 \Rightarrow$ con eliptic, cu dreptoldere „în sus"



$\sqrt{2x^2 + y^2} \leq z \leq \sqrt{1 - 2x^2 - y^2}$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_\Delta \left(\int_{\sqrt{2x^2+y^2}}^{\sqrt{1-2x^2-y^2}} dz \right) dx\,dy = \iint_\Delta \left(\sqrt{1-2x^2-y^2} - \sqrt{2x^2+y^2} \right) dx\,dy$$

$2x^2 + y^2 = 1 - 2x^2 + y^2 \iff 4x^2 + 2y^2 = 1 \iff \frac{x^2}{\left(\frac{1}{2}\right)^2} + \frac{y^2}{\left(\frac{1}{\sqrt{2}}\right)^2} = 1 \Rightarrow \Delta$ este interiorul unei elipse de semiaxe $a = \frac{1}{2}$, $b = \frac{1}{\sqrt{2}}$

$\Delta = pr_{xOy}\Omega = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; \frac{x^2}{\left(\frac{1}{2}\right)^2} + \frac{y^2}{\left(\frac{1}{\sqrt{2}}\right)^2} \leq 1 \right\}$

$x = \frac{1}{2} r \cos t$

$y = \frac{1}{\sqrt{2}} r \sin t$

$J(r,t) = \frac{1}{2\sqrt{2}} r \Rightarrow dx\,dy = \frac{1}{2\sqrt{2}} r\,dr\,dt$

$r \in [0,1], \; t \in [0, 2\pi]$

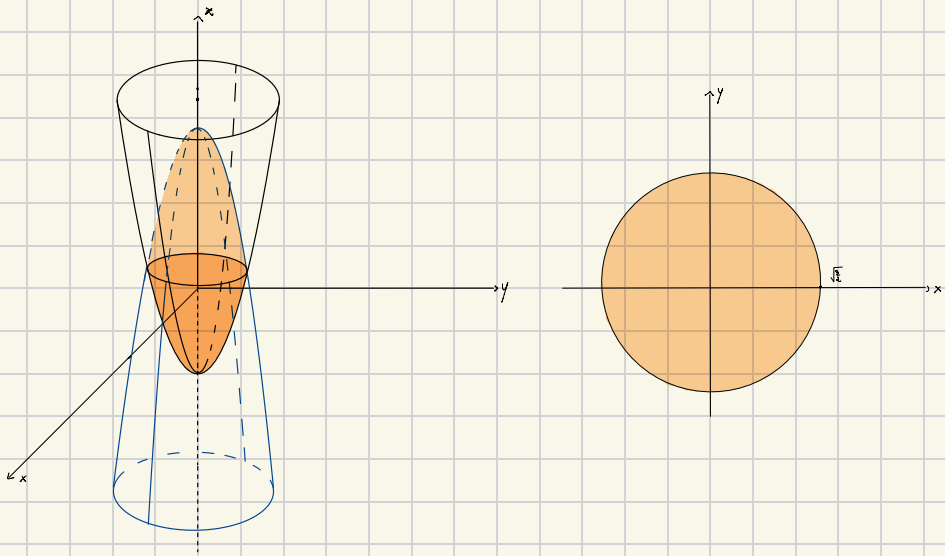
$$\iint_\Delta \left(\sqrt{1-2x^2-y^2} - \sqrt{2x^2+y^2} \right) dx\,dy = \int_0^{2\pi} \left(\int_0^1 \left(\frac{1}{2\sqrt{2}} r \sqrt{1 - \frac{1}{2} r^2} - \frac{1}{4} r^2 \right) dr \right) dt = \int_0^{2\pi} \frac{-1+\sqrt{2}}{6}\,dt = \frac{\pi(\sqrt{2}-1)}{3} \Rightarrow V = \frac{\pi(\sqrt{2}-1)}{3}$$

$$\int_0^1 \frac{1}{2\sqrt{2}} r \sqrt{1 - \frac{1}{2} r^2}\,dr = -\frac{1}{2\sqrt{2}} \int_0^1 \left(1 - \frac{1}{2} r^2\right)' \sqrt{1 - \frac{1}{2} r^2}\,dr = -\frac{1}{3\sqrt{2}} \left(1 - \frac{1}{2} r^2\right)^{\frac{3}{2}} \Big|_0^1 = -\frac{1}{12} + \frac{\sqrt{2}}{6}$$

6. $z = x^2 + y^2 - 1$, $z = 2 - x^2 - y^2$

$z = x^2 + y^2 - 1 \iff x^2 + y^2 = z + 1 \Rightarrow$ paraboloid eliptic cu vârful în $(0,0,-1)$, deschidere „în sus", rectiune cere, axă de simetrie Oz

$z = 2 - x^2 - y^2 \iff x^2 + y^2 = -z + 2 = \frac{z-2}{-1} \Rightarrow$ paraboloid eliptic cu vârful în $(0,0,2)$, deschidere „în jos", rectiune cere, axă de simetrie Oz.



$x^2 + y^2 - 1 \le z \le 2 - x^2 - y^2$

$$V = \iiint_{\Omega} dx\,dy\,dz = \iint_{\Delta} \left(\int_{x^2+y^2-1}^{2-x^2-y^2} dz \right) dx\,dy = \iint_{\Delta} (3 - 2x^2 - 2y^2)\,dx\,dy$$

$x^2 + y^2 - 1 = 2 - x^2 - y^2 \iff 2x^2 + 2y^2 = 3 \iff x^2 + y^2 = \left(\sqrt{\frac{3}{2}}\right)^2 \Rightarrow \Delta$ este interiorul unui cerc de rază $\sqrt{\frac{3}{2}}$ și centru $(0,0)$

$\Delta = pr_{xOy}\,\Omega = \left\{ (x,y) \in \mathbb{R}^2 \,\middle|\, x^2 + y^2 \le \left(\sqrt{\frac{3}{2}}\right)^2 \right\}$

$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r \in \left[0, \sqrt{\frac{3}{2}}\right]$, $t \in [0, 2\pi]$

$$\iint_{\Delta} (3 - 2x^2 - 2y^2)\,dx\,dy = \int_0^{2\pi} \left(\int_0^{\sqrt{\frac{3}{2}}} (3r - 2r^3)\,dr \right) dt = \frac{9}{8} \int_0^{2\pi} dt = \frac{9}{4}\pi \Rightarrow V = \frac{9}{4}\pi$$

7. $z = 4 - x^2 - y^2$, $2z = 5 + x^2 + y^2$

$z = 4 - x^2 - y^2 \iff x^2 + y^2 = \frac{z-4}{-1} \Rightarrow$ paraboloid eliptic cu vârful în $(0,0,4)$, dreptele „în jos", secțiune cercuri de rază 1, axă de simetrie Oz

$2z = 5 + x^2 + y^2 \iff x^2 + y^2 = \frac{z - \frac{5}{2}}{\frac{1}{2}} \Rightarrow$ paraboloid eliptic cu vârful în $(0,0,\frac{5}{2})$, dreptele „în sus", secțiune cercuri de rază 1, axă de simetrie Oz

Desenul este asemănător cu cel de la exercițiul **7.**

$\frac{1}{2}(5 + x^2 + y^2) \le z \le 4 - x^2 - y^2$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{\frac{1}{2}(5+x^2+y^2)}^{4-x^2-y^2} dz \right) dx\,dy = \frac{3}{2} \iint_D (1 - x^2 - y^2)\,dx\,dy$$

$4 - x^2 - y^2 = \frac{5}{2} + \frac{1}{2}x^2 + \frac{1}{2}y^2 \Rightarrow \frac{3}{2}x^2 + \frac{3}{2}y^2 = \frac{3}{2} \Rightarrow x^2 + y^2 = 1 \Rightarrow$ D este interiorul unui cerc de centru $(0,0)$ și rază 1

$D = pr_{xOy}\Omega = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1\}$

$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

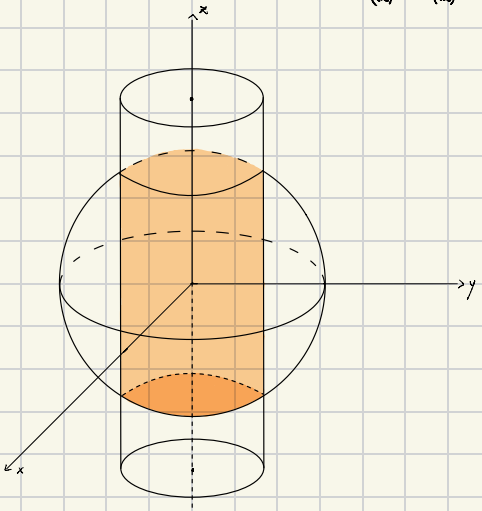
$r \in [0,1]$, $t \in [0, 2\pi]$

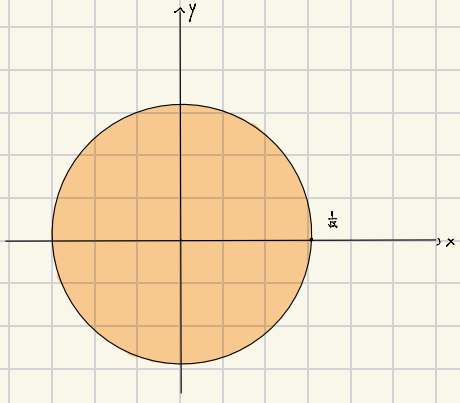
$$\frac{3}{2} \iint_D (1 - x^2 - y^2)\,dx\,dy = \frac{3}{2} \int_0^{2\pi} \left(\int_0^1 (r - r^3)\,dr \right) dt = \frac{3}{8} \int_0^{2\pi} dt = \frac{3}{4}\pi \Rightarrow V = \frac{3}{4}\pi$$

8. $x^2 + y^2 + z^2 = 1$, $x^2 + y^2 = \frac{1}{2}$

$x^2 + y^2 + z^2 = 1 \Rightarrow$ sferă centrată în origine, de rază 1

$x^2 + y^2 = \frac{1}{2} \Rightarrow 2x^2 + 2y^2 - 1 = 0 \Rightarrow \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + \frac{y^2}{\left(\frac{1}{\sqrt{2}}\right)^2} - 1 = 0 \Rightarrow$ cilindru circular drept, secțiune cerc de rază $\frac{1}{\sqrt{2}}$, axă de simetrie Oz.





Vom calcula volumul mulțimii formate „în interiorul" cilindrului și mărginite de cele două calote ale sferei.

$-\sqrt{1-x^2-y^2} \le z \le \sqrt{1-x^2-y^2}$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_D \left(\int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz \right) dx\,dy = 2 \iint_D \sqrt{1-x^2-y^2}\, dx\,dy$$

$D = pr_{xOy}\,\Omega = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; x^2+y^2 \le \frac{1}{2} \right\}$

$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r \in \left[0, \frac{1}{\sqrt{2}}\right]$, $t \in [0, 2\pi]$

$$2 \iint_D \sqrt{1-x^2-y^2}\, dx\,dy = \int_0^{2\pi} \left(\int_0^{1/\sqrt{2}} 2r\sqrt{1-r^2}\, dr \right) dt = \frac{4-\sqrt{2}}{6} \int_0^{2\pi} dt = \frac{\pi(4-\sqrt{2})}{3}$$

Volumul mulțimii formate în „exteriorul" cilindrului și în „interiorul" sferei este egal cu diferența dintre volumul sferei și volumul calculat anterior.

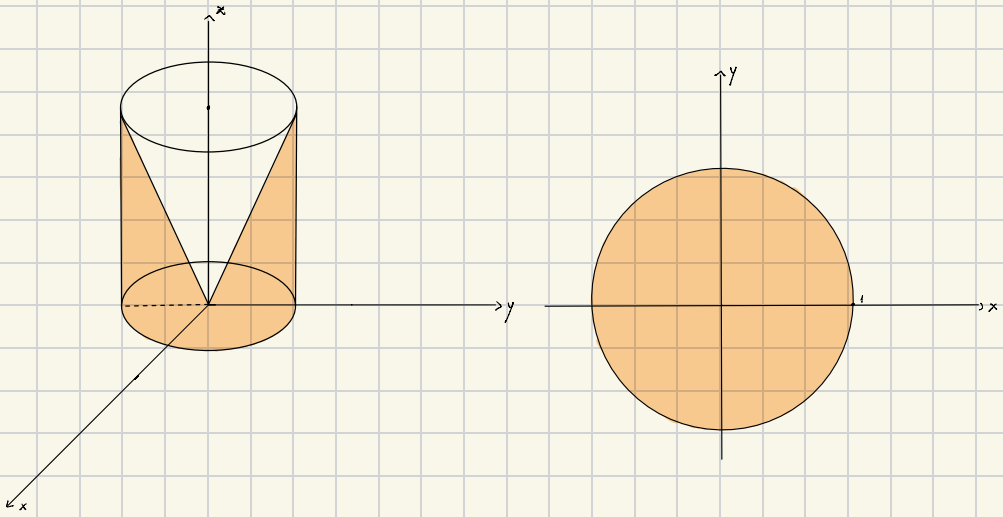
$$V^* = V_{sferă} - V = \frac{4\pi}{3} - \frac{4\pi - \sqrt{2}\pi}{3} = \frac{\sqrt{2}}{3}\pi$$

9.

$x^2 + y^2 = 1$, $z^2 = x^2 + y^2$, $z \ge 0$

$x^2 + y^2 = 1 \Rightarrow$ cilindru circular drept, secțiune cerc de rază 1, axă de simetrie Oz

$z^2 = x^2 + y^2 \Leftrightarrow x^2 + y^2 - z^2 = 0 \Rightarrow$ con elyptic, cu secțiune cerc, deschidere „în sus"



$0 \le z \le \sqrt{x^2+y^2}$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_\Delta \left(\int_0^{\sqrt{x^2+y^2}} dz \right) dx\,dy = \iint_\Delta \sqrt{x^2+y^2}\,dx\,dy$$

$\Delta = pr_{xOy}\,\Omega = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1\}$

$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r \in [0,1]$, $t \in [0,2\pi]$

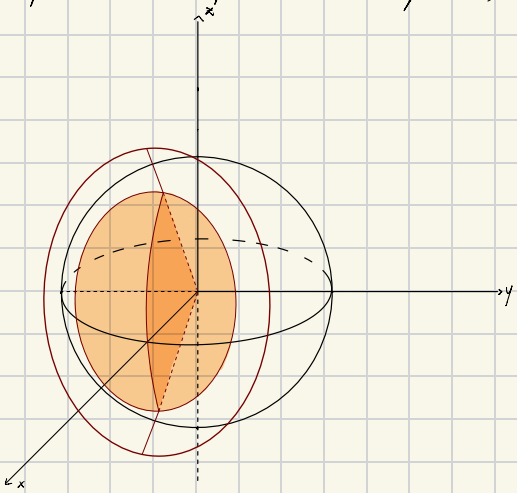
$$\iint_\Delta \sqrt{x^2+y^2}\,dx\,dy = \int_0^{2\pi} \left(\int_0^1 r^2\,dr \right) dt = \frac{1}{3}\int_0^{2\pi} dt = \frac{2\pi}{3}$$

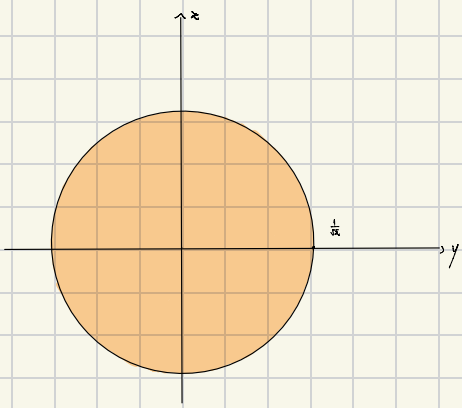
10.

$x^2+y^2+z^2 = 1$, $y^2+z^2 = x^2$, $x \ge 0$

$x^2+y^2+z^2 = 1 \Rightarrow$ sferă centrată în origine, de rază 1

$y^2+z^2 = x^2 \Leftrightarrow -x^2+y^2+z^2 = 0 \Rightarrow$ con eliptic, secțiune sferă, dischidere pe axa Ox





$\sqrt{y^2+z^2} \le x \le \sqrt{1-y^2-z^2}$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_\Delta \left(\int_{\sqrt{y^2+z^2}}^{\sqrt{1-y^2-z^2}} dx \right) dy\,dz = \iint_\Delta \left(\sqrt{1-y^2-z^2} - \sqrt{y^2+z^2} \right) dy\,dz$$

Pentru a obține $\Delta = pr_{yOz}\,\Omega$, îl „eliminăm" pe x din ecuațiile ecuațiilor $\Rightarrow y^2+z^2 = 1-y^2+z^2 \Leftrightarrow y^2+z^2 = \left(\frac{1}{\sqrt{2}}\right)^2$

$\Delta = pr_{yOz}\,\Omega = \left\{(y,z) \in \mathbb{R}^2 \,\middle|\, y^2+z^2 \le \left(\frac{1}{\sqrt{2}}\right)^2\right\}$

$y = r\cos t$

$z = r\sin t$

$J(r,t) = r \Rightarrow dy\,dz = r\,dr\,dt$

$r \in \left[0, \frac{1}{\sqrt{2}}\right]$, $t \in [0, 2\pi]$

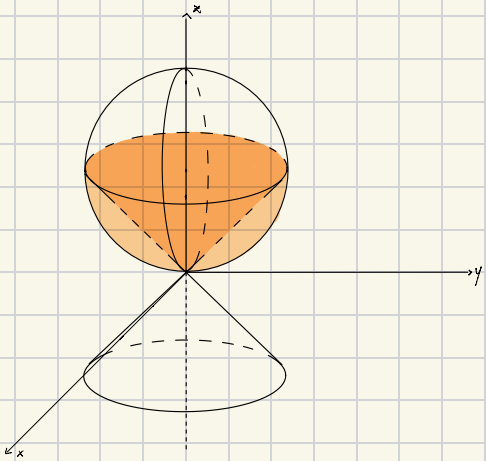
$$\iint_b \left(\sqrt{1-y^2-z^2} - \sqrt{y^2+z^2}\right) dy\,dz = \int_0^{2\pi}\left(\int_0^{1/\sqrt{2}} \left(r\sqrt{1-r^2} - r^2\right) dr\right) dt = \frac{2-\sqrt{2}}{6}\int_0^{2\pi} dt = \frac{(2-\sqrt{2})\pi}{3}$$

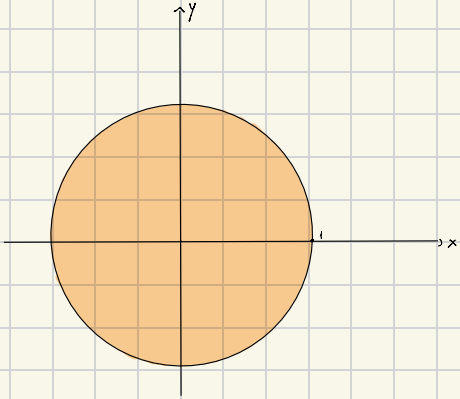
11.

$x^2 + y^2 = z^2$, $x^2 + y^2 + z^2 = 2z$

$x^2 + y^2 = z^2 \iff x^2 + y^2 - z^2 = 0 \Rightarrow$ con elliptic, secțiuni cercuri, dreptele pe care se află Oz

$x^2 + y^2 + z^2 = 2z \iff x^2 + y^2 + (z-1)^2 = 1 \Rightarrow$ sferă centrată în $(0,0,1)$, de rază 1.





$1 - \sqrt{1-x^2-y^2} \le z \le \sqrt{x^2+y^2}$

$$V = \iiint_\Omega dx\,dy\,dz = \iint_b \left(\int_{1-\sqrt{1-x^2-y^2}}^{\sqrt{x^2+y^2}} dz\right) dx\,dy = \iint_b \left(\sqrt{x^2+y^2} - 1 + \sqrt{1-x^2-y^2}\right) dx\,dy$$

$b = pr_{xOy}\,\Omega = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1\}$

$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r \in [0,1]$, $t \in [0, 2\pi]$

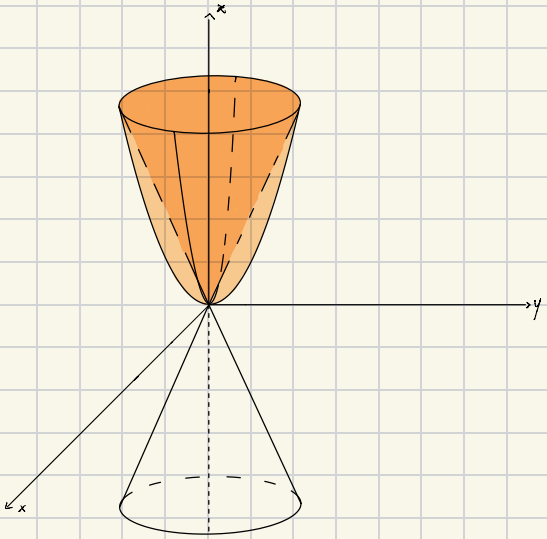
$$\iint_b \left(\sqrt{x^2+y^2} - 1 + \sqrt{1-x^2-y^2}\right) dx\,dy = \int_0^{2\pi}\left(\int_0^1 \left(r^2 - r + r\sqrt{1-r^2}\right) dr\right) dt = \frac{1}{2}\int_0^{2\pi} dt = \frac{\pi}{3}$$

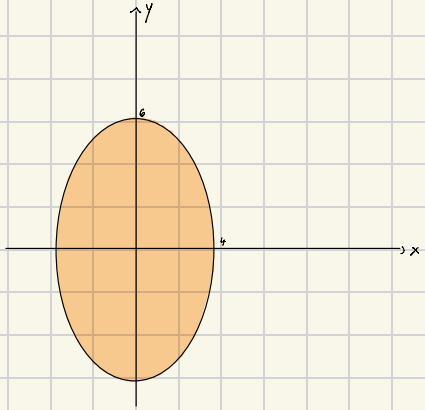
12.

$\frac{x^2}{4} + \frac{y^2}{9} = 2z$, $\frac{x^2}{4} + \frac{y^2}{9} = z^2$

$\frac{x^2}{4} + \frac{y^2}{9} = \frac{z}{\frac{1}{2}}$ $\Rightarrow$ paraboloid eliptic cu vârful în origine , descluderea „în sus", axeă de simetrie Oz

$\frac{x^2}{4} + \frac{y^2}{9} - z^2 = 0$ $\Rightarrow$ con eliptic, cu descluderea pe axeă Oz.





$\frac{x^2}{9} + \frac{y^2}{16} \le z \le \sqrt{\frac{x^2}{4} + \frac{y^2}{9}}$

$$V = \iiint_{\Omega} dx\,dy\,dz = \iint_{A} \left(\int_{\frac{x^2}{9} + \frac{y^2}{16}}^{\sqrt{\frac{x^2}{4} + \frac{y^2}{9}}} dz \right) dx\,dy = \iint_{A} \left(\sqrt{\frac{x^2}{4} + \frac{y^2}{9}} - \frac{x^2}{9} - \frac{y^2}{16} \right) dx\,dy$$

$A = pr_{xOy}\,\Omega = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; \frac{x^2}{16} + \frac{y^2}{36} \le 1 \right\}$

$x = 4r\cos t$

$y = 6r\sin t$

$r \in [0,1]$, $t \in [0, 2\pi]$

$J(r,t) = 24r \Rightarrow dx\,dy = 24r\,dr\,dt$

$$\iint_{A} \left(\sqrt{\frac{x^2}{4} + \frac{y^2}{9}} - \frac{x^2}{9} - \frac{y^2}{16} \right) dx\,dy = \int_0^{2\pi} \left(\int_0^1 24r(2r - 2r^2)\,dr \right) dt = 8\pi$$

13. $(x^4+y^4+z^4)^5 = x$

Facciamo la coordinate sferiche: $x = r\sin\theta\cos\varphi$

$y = r\sin\theta\sin\varphi$, $r>0$, $\theta\in[0,\pi]$, $\varphi\in[0,2\pi]$

$z = r\cos\theta$

$J(r,\theta,\varphi) = r^2\sin\theta$

$(x^4+y^4+z^4)^5 = x \iff r^8 = r\sin\theta\cos\varphi \implies \sin\theta\cos\varphi \geq 0 \implies \theta\in[0,\pi]$, $\varphi\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$\Downarrow$

$r = \sqrt[8]{\sin\theta\cos\varphi} \implies r\in[0,\sqrt[8]{\sin\theta\cos\varphi}]$

$$V = \int_0^{\pi}\int_{-\pi/2}^{\pi/2}\left(\int_0^{\sqrt[8]{\sin\theta\cos\varphi}} r^2\sin\theta\,dr\right)d\varphi\,d\theta = \frac{1}{3}\int_0^{\pi}\left(\int_{-\pi/2}^{\pi/2}\sin^{\frac{1}{2}}\theta\cos^{\frac{3}{8}}\varphi\,d\varphi\right)d\theta = \frac{1}{3}\int_0^{\pi}\sin^{\frac{1}{2}}\theta\left(\int_{-\pi/2}^{\pi/2}\cos^{\frac{3}{8}}\varphi\,d\varphi\right)d\theta$$

$$\int_{-\pi/2}^{\pi/2}\cos^{\frac{3}{8}}\varphi\,d\varphi = 2\int_0^{\pi/2}\cos^{\frac{3}{8}}\varphi\,d\varphi = \int_0^1(1-y)^{-\frac{1}{2}}y^{-\frac{1}{2}}\,dy = B\left(\tfrac{5}{8},\tfrac{1}{2}\right) = \frac{\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{13}{10}\right)}$$

$\sin^2\varphi = y \implies \sin\varphi = \sqrt{y} \implies \cos\varphi\,d\varphi = \frac{1}{2\sqrt{y}}\,dy$

$\varphi = 0 \implies y = 0$

$\varphi = \frac{\pi}{2} \implies y = 1$

$$\frac{1}{3}\int_0^{\pi}\sin^{\frac{1}{2}}\theta\left(\int_{-\pi/2}^{\pi/2}\cos^{\frac{3}{8}}\varphi\,d\varphi\right)d\theta = \frac{1}{3}B\left(\tfrac{5}{8},\tfrac{1}{2}\right)\cdot\int_0^{\pi}\sin^{\frac{1}{2}}\theta\,d\theta = \frac{1}{3}B\left(\tfrac{5}{8},\tfrac{1}{2}\right)\cdot B\left(\tfrac{1}{2},\tfrac{13}{10}\right) = \frac{1}{3}\cdot\frac{\Gamma\left(\frac{5}{8}\right)\cdot\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{13}{10}\right)}\cdot\frac{\Gamma\left(\frac{13}{10}\right)\cdot\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{9}{8}\right)} = \frac{1}{3}\cdot\frac{5\pi}{4} = \frac{5\pi}{12}$$

$$\int_0^{\pi}\sin^{\frac{1}{2}}\theta\,d\theta = \int_0^{\pi/2}\sin^{\frac{1}{2}}\theta\,d\theta + \int_{\pi/2}^{\pi}\sin^{\frac{1}{2}}\theta\,d\theta = \int_0^1 u^{-\frac{1}{4}}(1-u)^{\frac{3}{10}}\,du = B\left(\tfrac{1}{2},\tfrac{13}{10}\right)$$

$\cos^2\theta = u \implies \cos\theta = \sqrt{u} \implies \sin\theta\,d\theta = -\frac{1}{2\sqrt{u}}\,du$

$\theta = 0 \implies u = 1$

$\theta = \frac{\pi}{2} \implies u = 0$

$\cos^2\theta = u \implies \cos\theta = -\sqrt{u} \implies \sin\theta\,d\theta = \frac{1}{2\sqrt{u}}\,du$

$\theta = \frac{\pi}{2} \implies u = 0$

$\theta = \pi \implies u = 1$

14. $(x^2+y^2+z^2)^2 = 8x$

Facciamo la coordinate sferiche: $x = r\sin\theta\cos\varphi$

$y = r\sin\theta\sin\varphi$, $r>0$, $\theta\in[0,\pi]$, $\varphi\in[0,2\pi]$

$z = r\cos\theta$

$J(r,\theta,\varphi) = r^2\sin\theta$

$(x^2+y^2+z^2)^2 = 8x \iff r^4 = 8r\sin\theta\cos\varphi \implies \sin\theta\cos\varphi \geq 0 \implies \theta\in[0,\pi]$, $\varphi\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$\Downarrow$

$r = 2\sqrt[3]{\sin\theta\cos\varphi} \implies r\in[0,2\sqrt[3]{\sin\theta\cos\varphi}]$

$$V = \int_0^{\pi}\int_{-\pi/2}^{\pi/2}\left(\int_0^{2\sqrt[3]{\sin\theta\cos\varphi}} r^2\sin\theta\,dr\right)d\varphi\,d\theta = \frac{8}{3}\int_0^{\pi}\left(\int_{-\pi/2}^{\pi/2}\sin^2\theta\cos\varphi\,d\varphi\right)d\theta = \frac{16}{3}\int_0^{\pi}\sin^2\theta\,d\theta = \frac{16}{3}\int_0^{\pi}\frac{1-\cos 2\theta}{2}\,d\theta =$$

$$= \frac{8}{3}\pi - \frac{1}{4}\int_0^{\pi}2\cos 2\theta\,d\theta = \frac{8}{3}\pi - 0 = \frac{8}{3}\pi$$

15. $(x^2+y^2)^3 = z^2$, $z=0$, $z=8$

Fac trecerea la coordonatele cilindrice: $x = r\cos t$

$y = r\sin t$, $r>0$, $t \in [0, 2\pi]$

$z = z$

$J(r,t,z) = r$

$(x^2+y^2)^3 = z^2 \Rightarrow r^6 = z^2 \Rightarrow r = \sqrt[3]{z} \Rightarrow r \in [0,2]$, $t \in [0, 2\pi]$

$$V = \int_0^{2\pi}\int_0^{2}\left(\int_0^{8} r\,dz\right)dr\,dt = 8\int_0^{2\pi}\left(\int_0^{2} r\,dr\right)dt = 16\int_0^{2\pi} dt = 32\pi$$

16. $\frac{x^2}{a^2}+\frac{y^2}{b^2} = \frac{z^2}{c^2}$, $0 \le z \le c$, $a, b, c > 0$

$\frac{x^2}{a^2}+\frac{y^2}{b^2} = \frac{z^2}{c^2} \Leftrightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2} = 0 \Rightarrow$ con elipdic, descludere pe axa Oz

$x = ar\cos t$

$y = br\sin t$

$z = z$

$J(r,t,z) = abr$

$\frac{x^2}{a^2}+\frac{y^2}{b^2} = \frac{z^2}{c^2} \Rightarrow r^2 = \frac{z^2}{c^2} \Rightarrow r = \frac{z}{c} \Rightarrow r \in \left[0, \frac{z}{c}\right]$, $t \in [0, 2\pi]$

$$V = \int_0^{2\pi}\int_0^{c}\left(\int_0^{z/c} abr\,dr\right)dz\,dt = \frac{\pi}{3}abc$$